Possible Causes and Solutions

1. Missing Values

Possible Causes:

- Customers may not have provided all the required details while placing their orders.
- Payment or discount information might have been missed during data entry.
- Some information may have been lost while importing data from another system.
- Human errors can also lead to incomplete records.

Solutions:

- Make important fields mandatory when customers place orders.
- Check the original records to recover missing information whenever possible.
- If the actual value cannot be found, use a suitable placeholder such as "Unknown" for missing city or payment details.
- Fill missing numerical values only when there is a reasonable basis for doing so.
- Regularly check the dataset for missing values to prevent problems during analysis.

2. Duplicate Records

Possible Causes:

- The same data might have been imported more than once.
- A customer may have submitted an order multiple times.
- Technical issues could have caused the same transaction to be recorded repeatedly.
- Errors in the system might have resulted in repeated order IDs.

Solutions:

- Check for duplicate records before starting the analysis.
- Make sure every order has a unique order ID.
- Verify suspected duplicates using the original order details before removing them.
- Use Pandas functions such as `duplicated()` to identify repeated records.
- Avoid deleting records without checking them, as two identical-looking orders may belong to separate purchases.

3. Group-Wise Data Analysis

Possible Causes:

- Some cities may have more customers and orders than others.
- Customers in different cities may purchase different quantities or products.

- Discounts, product availability, and local demand can affect revenue.
- Missing or incorrectly entered city names may lead to inaccurate city-wise results.

Solutions:

- Compare the number of orders, total quantity sold, total revenue, and average order value for each city.
- Check whether differences in revenue are caused by more orders or higher spending per order.
- Plan stock levels according to customer demand in each city.
- Offer suitable promotions in cities where sales are lower, after investigating the reasons.
- Correct inconsistent city names and investigate missing location details before grouping the data.
- Use a larger dataset before making major business decisions based on the results.

4. Identifying the Highest-Priced Item

Possible Causes:

- Some products may naturally have higher prices because of their edition, quality, or supplier cost.
- A price may have been entered incorrectly.
- Currency symbols or text values may have caused problems during data processing.
- Invalid entries, such as `"invalid"` in the price column, may prevent accurate comparisons.

Solutions:

- Convert the price column into a numerical format before finding the highest price.
- Check invalid price entries against the original product records.
- Verify unusually high prices before deciding whether they are correct.
- Use Pandas functions such as `max()`, `idxmax()`, and `nlargest()` to identify the highest-priced items.
- Add validation rules to prevent incorrect prices from being entered into the system.
- Compare product prices with supplier costs and market prices before making pricing decisions.